Answers learning task 11

2026 – 2027

Exercise 7.21: Let $X_1, X_2, \ldots, X_7$ denote a random sample from a population having mean μ and variance σ^2. Consider the following estimators of μ:

$$\hat{\Theta}_1 = \frac{X_1 + X_2 + \cdots + X_7}{7}$$

$$\hat{\Theta}_2 = \frac{2X_1 - X_6 + X_4}{2}$$

(a) Is either estimator unbiased?

(b) Which estimator is better? In what sense is it better? Calculate the relative efficiency of the two estimators.

Answer

(a) Both estimators are unbiased because $E(\hat{\Theta}_1) = \mu$ and $E(\hat{\Theta}_2) = \mu$.

(b) $\hat{\Theta}_1$ is better because it is more precise. Since $V(\hat{\Theta}_1) = \frac{\sigma^2}{7}$ and $V(\hat{\Theta}_2) = \frac{3\sigma^2}{2}$, the relative efficiency of $\hat{\Theta}_2$ to $\hat{\Theta}_1$ is $\frac{2}{21}$.

Exercise 7.23: Suppose that $\hat{\Theta}_1$ and $\hat{\Theta}_2$ are estimators of the parameter θ. We know that $E(\hat{\Theta}_1) = \theta$, $E(\hat{\Theta}_2) = \theta/2$, $V(\hat{\Theta}_1) = 10$, $V(\hat{\Theta}_2) = 5$. Which estimator is better? In what sense is it better?

Answer

$\hat{\Theta}_1$ is preferred. $\hat{\Theta}_1$ is unbiased with $V(\hat{\Theta}_1) = 10$. $\hat{\Theta}_2$ is biased, but $2\hat{\Theta}_2$ is unbiased with $V(2\hat{\Theta}_2) = 20$. Since $\hat{\Theta}_1$ has a smaller variance among the unbiased options, it is the better estimator.

Exercise 7.25: Let three random samples of sizes $n_1 = 8$, $n_2 = 14$, and $n_3 = 6$ be taken from a population with mean μ and variance σ^2. Let S_1^2, S_2^2, and S_3^2 be the sample variances. Show that $S^2 = (7S_1^2 + 13S_2^2 + 5S_3^2)/25$ is an unbiased estimator of σ^2.

Answer

Since $E(S_1^2) = E(S_2^2) = E(S_3^2) = \sigma^2$, we have:

$$E(S^2) = \frac{7}{25}\sigma^2 + \frac{13}{25}\sigma^2 + \frac{5}{25}\sigma^2 = \sigma^2.$$

Thus, S^2 is an unbiased estimator of σ^2.

Exercise 7.29: Data on the oxide thickness of semiconductor wafers are as follows: 425, 431,

416, 420, 421, 436, 418, 410, 431, 433, 423, 426, 411, 435, 436, 428, 411, 426, 407, 437, 422, 428, 413, 416.

(a) Calculate a point estimate of the mean oxide thickness for all wafers in the population.

(b) Calculate a point estimate of the standard deviation of oxide thickness for all wafers in the population.

(c) Calculate the standard error of the point estimate from part (a).

(d) Calculate a point estimate of the median oxide thickness for all wafers in the population.

(e) Calculate a point estimate of the proportion of wafers in the population that have oxide thickness of more than 430 angstroms.

Answer

(a) 423.3333 (b) 9.1493 (c) 1.8676
(d) 424 (e) 0.2917

Exercise 7.32: Two different plasma etchers in a semiconductor factory have the same mean etch rate μ. However, machine 1 is newer than machine 2 and consequently has smaller variability in etch rate. We know that the variance of etch rate for machine 1 is σ_1^2, and for machine 2, it is $\sigma_2^2 = a\sigma_1^2$. Suppose that we have n_1 independent observations on etch rate from machine 1 and n_2 independent observations on etch rate from machine 2.

(a) Show that $\hat{\mu} = \alpha\bar{X}_1 + (1-\alpha)\bar{X}_2$ is an unbiased estimator of μ for any value of α between zero and one.

(b) Find the standard error of the point estimate of μ in part (a).

(c) What value of α would minimize the standard error of the point estimate of μ?

(d) Suppose that $a = 4$ and $n_1 = 0.5n_2$. What value of α would you select to minimize the standard error of the point estimate of μ? How "bad" would it be to arbitrarily choose $\alpha = 0.5$ in this case?

Answer

(a) $E(\hat{\mu}) = \alpha E(\bar{X}_1) + (1-\alpha)E(\bar{X}_2) = \alpha\mu + (1-\alpha)\mu = \mu$

(b) S.E. $= \sigma_1\sqrt{\frac{\alpha^2}{n_1} + \frac{a(1-\alpha)^2}{n_2}}$

(c) $\alpha = \frac{an_1}{n_2+an_1}$

(d) $\alpha = \frac{2}{3}$, choosing $\alpha = 0.5$ increases the standard error and is thus not preferable.